elements of an operator, assuming that we already know what the operator does to each basis vector.

Exercise 7.1: Write the tensor product $I \otimes \tau_x$ as a matrix, and apply that matrix to each of the $|uu\rangle$, $|ud\rangle$, $|du\rangle$, and $|dd\rangle$ column vectors. Show that Alice's half of the state-vector is unchanged in each case. Recall that I is the 2×2 unit matrix.

Exercise 7.2: Calculate the matrix elements of $\sigma_z \otimes \tau_x$ by forming inner products as we did in Eq. 7.2.

The third exercise is a bit tedious, but it really nails things down. Consider the equation

$$(A \otimes B)\ \ (a \otimes b) = (Aa \otimes Bb). \qquad (7.10)$$

As in Eqs. 7.7 and 7.8, A and B represent 2×2 matrices (or operators), and a and b represent 2×1 column vectors. The exercise asks you to expand the equation into components and show that the left side matches the right side.

Exercise 7.3:

a) Rewrite Eq. 7.10 in component form, replacing the symbols A, B, a, and b with the matrices and column vectors from Eqs. 7.7 and 7.8.

b) Perform the matrix multiplications Aa and Bb on the right-hand side. Verify that each result is a 4×1 matrix.

c) Expand all three Kronecker products.

d) Verify the row and column sizes of each Kronecker product: